

IBRAHEEM OBANLA

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with five years of experience building and deploying models for credit risk, customer retention and security analytics. Statistics-trained, with published research applying attention-based deep learning to early Parkinson's disease detection. Experienced across the full model lifecycle, from data pipelines and feature engineering through to deployment and monitoring, working in Python, R and SQL on AWS and Azure.

TECHNICAL SKILLS

Languages: Python, R, SQL

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, SHAP, Ensemble Methods, Feature Engineering

NLP: NLTK, spaCy

Cloud & MLOps: AWS (SageMaker, EC2, S3), Azure, Model Monitoring, Version Control

Data & Visualisation: pandas, NumPy, Power BI, Tableau, Excel, ETL & Data Pipeline Development

Deployment: Flask, Streamlit, FastAPI

Statistics: Hypothesis Testing, A/B Testing, Regression Modelling, Statistical Inference

PROFESSIONAL EXPERIENCE

TechSTEPS Limited | Lagos, Nigeria

Machine Learning Engineer | June 2023 – July 2026

- Spearheaded the design and deployment of predictive analytics models using Python, R and AWS, proactively identifying at-risk customers and driving a 13% reduction in customer attrition.
- Supported deployment of machine learning models to Azure environments, ensuring performance monitoring and version control through MLOps practices.
- Led the development of automated reporting solutions using Power BI and SQL, reducing manual reporting effort and saving approximately 540 operational hours annually.
- Improved data accuracy and reliability across business-critical dashboards by implementing robust data quality checks, driving a 29% reduction in reporting errors and strengthening executive decision-making.

Armstrong Global Concepts LLC | Austin, TX

Data Scientist | February 2021 – January 2023

- Developed machine learning models with Scikit-learn and TensorFlow across 10,000,000+ customer records, uncovering insights that supported risk mitigation and customer retention initiatives.
- Analysed employee feedback using natural language processing in Python, NLTK and spaCy, identifying factors affecting workplace satisfaction and contributing to a 34% reduction in employee turnover.
- Designed and developed interactive Excel and Power BI dashboards visualising key HR and operational metrics, giving leadership the basis for data-informed retention strategies.
- Extracted, cleaned and transformed large datasets using SQL and Python, improving reporting efficiency by 41% and enhancing data reliability.

- Created reusable data pipelines with SQL and Python, improving efficiency for ongoing analysis and reporting.

SAMPLE PROJECTS

Credit Risk: Lending Club Default Prediction — github.com/tolatechy/Credit-Risk-Lending-Club

- Built default prediction models over 2.2M loans issued 2007 to 2018, comparing Logistic Regression, Decision Tree, Random Forest and XGBoost.
- Focused feature selection on eliminating post-outcome leakage, excluding fields such as recovery amounts that do not exist at the lending decision and inflate accuracy while failing in production.

Employee Attrition Prediction — github.com/tolatechy/Employee-Attrition-Prediction

- Modelled employee attrition across 1,470 records and 35 features, comparing Logistic Regression, Decision Tree, Random Forest and Gradient Boosting to a best cross-validated score of 0.907.
- Addressed class imbalance with SMOTE, as roughly one in six employees left and accuracy alone would have flattered a model that predicted retention for everyone.

Insider Threat Detection in Corporate Email — github.com/tolatechy/Insider-Threat-Email-Detection

- Classified corporate email traffic to surface insider threat behaviour, reaching 97.6% accuracy with a voting ensemble over Logistic Regression, Decision Tree and Random Forest, benchmarked against a Keras neural network.
- Handled severe class imbalance with SMOTE, since genuine insider activity is rare by definition.

Explainable Network Intrusion Detection — github.com/tolatechy/Intrusion-Detection-Explainable

- Integrated SHAP attribution into intrusion classifications so analysts can interrogate why traffic was flagged, deployed as a Streamlit application.
- Repurposed the same attribution analysis as a leakage diagnostic, since near-perfect benchmark scores on IDS datasets typically indicate separable labels rather than model quality.

Full portfolio of 13 projects: github.com/tolatechy

PUBLICATION

Attention-Based Deep Learning for Early Parkinson's Disease Detection with Tabular Biomedical Data
arxiv.org/abs/2602.07933

EDUCATION

University of Ibadan, Oyo State, Nigeria

Master of Science, Data and Information Science — 2026

University of Ibadan, Oyo State, Nigeria

Bachelor of Science, Statistics — 2021